

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1st Semester of M.Pharm (Clinical Pharmacy)

University Theory Examination December 2015

PH812.01 ANALYTICAL TECHNIQUES

Date: 16.12.2015, Wednesday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Explain following terms: Immunoassay, ELISA, Radio isotope, Northern Blotting Technique, Southern Blotting Technique. 05
- B) Write a note on different types of ELISA. 05
- Q.2 A) Write a note on principle of electrophoresis. 05
- B) Give the classification of electrophoresis and explain Zone electrophoresis. 05

OR

- Q.2 A) Write a note on flow cytometry. 05
- B) Write a note on Staining techniques for live cell imaging. 05
- Q.3 Answer the following (Any three) 15
- (A) What is Quenching? Explain it and differentiate fluorescence and phosphorescence.
- (B) Explain factors affecting the fluorescence intensity of a compound.
- (C) Write a note on Isoelectric Focussing (IEF).
- (D) Write a note on In-vivo cell tracking technique.

SECTION-II

- Q.4 (A) Give principle of NMR spectroscopy. How NMR spectroscopy helps in structure elucidation of compounds. 05
- (B) Define the terms 05
- (1) Mc-Lafferty rearrangement
- (2) Metastable peak
- Q.5 (A) Explain principle of UV-vis spectroscopy. 05
- (B) Give principle of IR spectroscopy. What is FT-IR. 05

OR

- Q.5 (A) Which detectors are used in Gas Chromatography. Explain working of any one detector. 05
- (B) Write a note on principle and instrumentation of super fluid chromatography. 05
- Q.6 Answer the following (Any three) 15
- (A) Explain plate theory for chromatography in details.
- (B) Write a brief note on Affinity chromatography.
- (C) Enlist the ionization techniques used in mass spectroscopy. Discuss MALDI techniques.
- (D) Write a note on HPTLC.